

Time-Frequency Analysis (ERSP)

1) Objectives of ESRP

- Show how brain wave power changes over time and across frequencies in response to events
- Compare post-event brain activity to a quiet baseline period before the event
- Reveal when and where different brain waves (Theta, Alpha, Beta, Gamma) get stronger or weaker
- Identify synchronization vs. desynchronization (brain coordination)

2) How It Works

ERSP (Event-Related Spectral Perturbation):

- Computes spectral power across time and frequency for each trial
- Compares post-event power to a pre-event baseline period
- Displays results as a **color-coded heat map** showing power changes in decibels (dB)
- Uses logarithmic frequency scaling to cover both low and high frequencies

Key outputs understood by researchers:

- **Timing** of brain responses (how fast/slow)
- **Frequency specificity** (which brain waves are affected)
- **Spatial patterns** (which brain areas respond)
- **Synchronization vs. desynchronization** (brain coordination)

Multi-Channel ERSP Analysis (Synthetic Events)
Frequency: 4-45 Hz | Baseline: -0.3-0 s | Window: -1-2 s

